Anand Bhattad

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EDUCATION

University of Illinois, Urbana-Champaign (UIUC), USA

Ph.D., Computer Science — *Jan'19 - Present*
Advisor: David Forsyth
Thesis committee: David Forsyth (chair), Alyosha Efros, Bill Freeman, Derek Hoiem, Shenlong Wang and Svetlana Lazebnik

M.S., Computer Science — *Aug'17 - Dec'18*
M.S., Civil Engineering — *Aug'15 - Aug'17*

National Institute of Technology Karnataka (NITK) Surathkal, India

B.Tech., Civil Engineering — *July'11 - May'15*

REFERENCES

David Forsyth, Professor, Computer Science, UIUC – daf@illinois.edu
Derek Hoiem, Professor, Computer Science, UIUC – dhoiem@illinois.edu
Shenlong Wang, Assistant Professor, Computer Science, UIUC – shenlong@illinois.edu
Yuxiong Wang, Assistant Professor, Computer Science, UIUC – yxw@illinois.edu

PREPRINTS /UNDER REVIEW

6. **StyLitGAN: Enriching StyleGAN with Illumination Physics**
A. Bhattad, D. A. Forsyth
In Submission

5. **Illuminator: Learning to Correct Lighting Without Ground Truth Data**
A. Bhattad, B.Chen, S. Wang, S. Richter, D. A. Forsyth
In Submission

4. **Make It So: Steering StyleGAN for Any Image Inversion and Editing**
A. Bhattad, V. Shah, D. Hoiem, D. A. Forsyth
In Submission

3. **Learning Equivariant Dense Prediction Networks**
Yuanyi Zhong, **A. Bhattad**, Y. Wang, D. A. Forsyth
In Submission

2. **Denoising Upsampling Residual Autoencoders: Easy MAP Estimate for Image Enhancement**
A. Bansal, **A. Bhattad**, D. A. Forsyth
In Submission

1. **SIRfyN: Single Image Relighting from your Neighbors**
D. A. Forsyth, **A. Bhattad**, P. Asthana, Y. Zhong, Y. Wang
Technical Report

CONFERENCE PAPERS

5. **Cut-and-Paste Object Insertion by Enabling Deep Image Prior for Reshading**
A. Bhattad, D. A. Forsyth
International Conference on 3D Vision (3DV), 2022

4. **DIVeR: Real-time and Accurate Neural Radiance Fields with Deterministic Integration for Volume Rendering** **(Best Paper Finalist)**
L. Wu J. Y. Lee **A. Bhattad** Y. Wang D. A. Forsyth
Computer Vision and Pattern Recognition (CVPR), 2022

3. **View Generalization for Single Image Textured 3D Models**
A. Bhattad, A. Dundar, G. Liu, A. Tao, B. Catanzaro
Computer Vision and Pattern Recognition (CVPR), 2021

2. **Unrestricted Adversarial Perturbations via Semantic Manipulation**
A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
International Conference on Learning Representations (ICLR) 2020

1. **Improved Style Transfer with Calibrated Metrics**
   M. Yeh*, S. Tang*, **A. Bhattad**, C. Zou, D. A. Forsyth (* for equal contribution)
   Winter Conference on Applications of Computer Vision (WACV) 2020

Workshop Papers

2. **Big but Imperceptible Adversarial Perturbations via Semantic Manipulation**
   A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
   Computer Vision and Pattern Recognition (CVPR) Workshop 2019, ***Oral Presentation***

1. **Detecting Anomalous Faces with "No Peeking" Autoencoders**
   A. Bhattad, J. Rock, D. A. Forsyth
   Computer Vision and Pattern Recognition (CVPR) Workshop, 2018, ***Oral Presentation***

Students Mentoring

- Matthew Weston (PhD; present) | Project on Controllable Diffusion Models
- James Soole (MS; present) | Project on Illumination Physics in Dense Prediction Models
- Feiran Wang (MEngg; present) | Project on NeRF + Illumination Physics
- Liwen Wu (BS-MS; 2021-2022) | Paper at CVPR 2022 | Next position: PhD Student at UCSD
- Brian Chen (BS; 2020-2022) | Paper under review | REU at CMU's Robotics Institute
- Kexuan (Klaus) Zou (MS; 2019-2020) | Next position: Software Engineer at NVIDIA
- Anchu Zhu (BS; 2018-2019) | Next position: MS in CS at USC

Research Internships

Allen Institute of Artificial Intelligence, Seattle/Champaign(remote) *Feb'23 - present*
Research Intern with *Ani Kembhavi*
Project on Unified generative-discriminative model.

Intel ISL, Santa Clara/Champaign (remote) *Feb'21 - Aug'21*
Research Intern with *Stephan R. Richter*
Project on image-based object insertion.

NVIDIA ADLR, Santa Clara/Champaign (remote) *May'20 - Dec'20*
Research Intern with *Aysegul Dundar, Andrew Tao and Bryan Catanzaro*
Project on single image textured 3D inference via differentiable rendering (CP#3)

Fyusion Inc, San Francisco *May'19 - Aug'19*
Research Intern with *Abhishek Kar and Rodrigo Ortiz Cayon*
Project on image-based relighting of cars.

Siemens Corporate Technology, Princeton *May'17 - Aug'17*
Computer Vision & DL Intern with *Jan Ernst and Ziyan Wu*
Developed a semi-automated segmentation tool based on Polygon-RNN from CVPR 2017.

Computational Sensor Lab, Michigan State University *May'14 - July'14*
Research Intern with *Prof. Nizar Lajnef and Prof. Amir H. Alavi*
Built a framework for Damage Detection Using Self-Powered Wireless Sensors

Earthquake Engineering Research Center, IIIT Hyderabad *Jun'13 - July'13*
Research Intern with *Prof. Ramacharla Pradeep Kumar*
Designed Earthquake Resistant Multi-Storied Building

Teaching Experience

Computer Science, UIUC; *Graduate Teaching Assistant* *Jan'16 - Present*

- Applied Machine Learning (CS498 AML) in Fall'18 | Designed Kaggle competitions
- Data Structures (CS 225) in Spring'17 | Developed Assignments, Exams & Labs
- Introduction to Computing (CS 101) in Spring'16 & Fall'16 | **Rated as Excellent TA**

Other Projects

Pedestrian Detection and Automated Braking for Self-Driving Cars *Jan'17 - May'17*
Project Mentor: Prof. David A. Forsyth `[Video | Code]`

- Built a smooth autonomous braking system for pedestrian detection using YOLO & SSD
- Estimated pedestrian distance from perspective geometry using only a monocular camera
- Integrated the software built using Robotics Operating System and tested on a self-driving car

Services & Leadership

Organizer, CVPR-W on "Big Models Vs Scholars: How Should Academics Adapt?" June 2023

- Co-organizing a CVPR workshop to discuss the potential impact of large vision models on the Academic CV community
- Aims to bring fresh perspectives on redesigning collaborations, access to compute, and curricula.
- Positively shape a more inclusive, equitable, and diversity-rich future for computer vision via discussions and opinions through assorted panels and speaker talks

Organizer, External Speaker Series and Vision Lunch Jan 2021-Present

- Hosting the first external speaker series, inviting researchers outside of UIUC for technical talks
- Organizing computer vision reading and presentation group in the CS department of UIUC

Reviewer, Computer Vision Conferences Nov 2018-Present

- Serving as reviewer for CVPR, NeurIPS, ECCV and ICCV conferences and workshops.
- **Outstanding Emergency Reviewer** for CVPR 2021

Program Committee Jun 2019

- Adversarial Machine Learning in Real-World Computer Vision Systems, CVPR 2019 Workshop
- Security and Privacy in Machine Learning, ICML 2019 Workshop

Organizer, Vision Coffee, Urbana-Champaign *May'18 - Aug'18*

- Vision and Deep Learning paper/topic weekly discussion group
- Finalized agenda/topic, location and initiated discussions

Co-Founder, SchoolEngg *Apr'13 - May'15*

- Forum for high-school students; providing the firsthand knowledge of all engineering disciplines
- Currently, renamed as *PrepLift*, an education counselling start-up in India